Application technical documentation

Little Learners and NurseryConnect

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Project versions and build environment

Project	Source and platform
Little Learners	Repository commit 744e1d2375b95f5b6e1a14bd7e39b79166bbde3a; iOS deployment target 17.0
NurseryConnect	Repository commit 2e3d8a2515585ecd7fd8be51f892397d0ef51928 plus the local SpatialDiaryView.swift update; visionOS deployment target 1.0

Both projects require a Mac with Xcode and the relevant simulator runtime to build and run. The Little Learners README specifies Xcode 26.2 or later, and the source uses newer Swift concurrency features. Use that project requirement rather than assuming the iOS deployment target identifies the required compiler version. Confirm the actual environment used in the final evidence.

Little Learners build instructions

• Clone or download https://github.com/ktthiveesan-glitch/LittleLearners.

• Open LittleLearners.xcodeproj in Xcode.

• Select the LittleLearners scheme and an installed iPhone or iPad simulator.

• Choose Product > Build, then Product > Run.

• For a physical device, set an appropriate signing team and bundle identifier in the target settings.

• Choose Product > Test to run the unit and UI targets. Save the result bundle or screenshots of the test report.

Command-line alternative from the repository root: xcrun simctl list devices available. Then run xcodebuild -project LittleLearners.xcodeproj -scheme LittleLearners -destination "platform=iOS Simulator,name=<installed simulator name>" -parallel-testing-enabled NO test. Replace the placeholder with the exact name returned by the first command.

NurseryConnect build instructions

• Clone or download https://github.com/ktthiveesan-glitch/MAAD-Assignment-Part-B.

• If demonstrating Add Entry, replace Views/SpatialDiaryView.swift with the updated version supplied in this pack before building.

• Open NurseryConnect.xcodeproj, select the NurseryConnect scheme and an installed Apple Vision Pro simulator.

• Choose Product > Build and Product > Run.

• Verify the saved entry appears and survives relaunch. Do not claim the local patch is tested until this succeeds.

The new AddDiaryEntryView is a private type inside SpatialDiaryView.swift. No new source-file reference or model migration is required for this patch. The repository on GitHub is not automatically updated by the local replacement.

Little Learners source organisation

Directory	Role
App	Application container, root screen, preview data and typed routes
Models	SwiftData child, progress, attempt and badge records
Content	Learning items, scripts, categories and badge definitions
ViewModels	Feature state and decision logic
Views	Onboarding, home, lessons, tracing, shapes, rewards and parent area
Services	Handwriting, guide generation, coverage, speech, progress and badges
Components and Theme	Reusable controls, animation modifiers and shared appearance
Resources and Assets.xcassets	String Catalog and colour assets
LittleLearnersTests and LittleLearnersUITests	Unit and UI test source

Data flow

LittleLearnersApp supplies a shared model container. RootView loads profiles and stores the active identifier using AppStorage. Router owns the navigation path. The selected ChildProfile is passed through the environment. Feature views use observable view models, which call services and update records linked to the active profile.

For tracing, normalised strokes become canvas points. HandwritingRecognizer chooses the Vision path for English or numbers and otherwise uses CoverageChecker. TracingViewModel records the attempt. A correct result adds a star, marks progress and checks badge eligibility. It then attempts to save through the profile's model context.

Dependencies and configuration

The project uses Apple frameworks including SwiftUI, SwiftData, Vision, AVFoundation, Charts and Core Text. The README declares no third-party packages. Language and sound preferences use AppStorage/UserDefaults. The inspected test launch mode uses an in-memory model container to keep UI tests separate from normal data.

Model deletion and progress reset must be tested explicitly. Exporting or syncing progress between devices is not implemented in the reviewed source.

NurseryConnect source organisation

File	Responsibility
NurseryConnectVisionApp.swift	Creates the NurseryConnectV2 SwiftData store and welcome/DiarySpace window groups; inserts samples
ContentView.swift	Welcome interface, openWindow action, entry count and attached keyworker ornament; also contains sample seeding
Models/DiaryEntry.swift	SwiftData record with UUID, childName, activityType, entryDescription, mood and date
Views/SpatialDiaryView.swift	Diary grid and detail selection; local patch adds Add Entry and newest-first sorting
Views/DiaryCardView.swift	Compact summary of one observation
Views/DiaryDetailView.swift	Full notes, child name, activity, mood and date in a sheet
Views/KeyworkerPanelView.swift	Entry count and current date ornament

Add Entry patch

The local update adds a sheet containing fields for the child's name, activity, mood and notes. Mood choices match the existing strings: Happy, Excited, Tired, Sad and Poorly. Save is enabled only when the trimmed name, activity and notes are non-empty. Cancel dismisses the form without inserting an entry.

On Save, the view constructs a DiaryEntry and inserts it into a separate ModelContext using the existing container, with autosave disabled. A successful explicit save dismisses the form. A failure rolls back that context and shows an alert while retaining the field text. This isolates rollback from changes elsewhere. Query refresh across contexts and windows must be verified in the simulator.

The diary query sorts by date descending. A scrollable adaptive grid replaces the previous two rows of three cards. The heading becomes Diary Entries because the query includes all dates. The model remains unchanged, including its automatic timestamp at construction.

Platform capabilities and limits

The app uses native visionOS window groups, openWindow/dismissWindow, an ornament and hover feedback. Cards are two-dimensional SwiftUI content inside a window. The reviewed source does not provide independent three-dimensional card placement, RealityKit content, an immersive space or a login flow. The keyworker role is a prototype label.

Known issues and maintenance priorities

Project	Issue	Recommended action
Little Learners	Silent save failures in several try? paths	Add visible errors and verify durable state
Little Learners	OCR mismatch can fall back to coverage	Document the behaviour and test wrong letters and scribbles
Little Learners	App icon images absent in inspected asset folder	Add an original icon and verify the build output
Little Learners	Translations await native-speaker review	Review the content in TRANSLATION_REVIEW.md
NurseryConnect	Sample insertion exists in two locations	Assign a single owner and handle fetch/save failures
NurseryConnect	Store initialisation uses fatalError	Provide a recoverable storage failure state
NurseryConnect	Keyworker panel displays SE4020	Correct the label to SE4041 before final recording
NurseryConnect	No implemented login or record access controls	Describe simulated role accurately; design authentication before real deployment
NurseryConnect	Add Entry patch has no executed build evidence	Build and exercise the workflow on a Mac

Reproducible verification

Record the commit, local patch status, Xcode version, target OS and simulator/device before testing. Capture build and test results from that same version. A recording made before applying Add Entry cannot demonstrate the new feature. Retain the unedited recording and label any edited highlights clearly.

For each failure, record the steps, expected result, actual result and evidence filename. Correct one cause at a time, rebuild and repeat the affected test. Keep performance measurements on a consistent device and describe whether they are cold launches or repeated launches.